NANDINI MEHTA

SOFTWARE ENGINEERING PORTFOLIO

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01 **Backend and APIs**

ChargeHolidays Hotel Booking Platform

Technologies: Go · REST APIs · PostgreSQL · Firebase · Elasticsearch · Kibana · Docker · Kubernetes · Google Cloud

Impact: Professional backend work on systems processing more than 50,000 events per day.

- Developed Go microservices and REST APIs backed by PostgreSQL and Firebase Firestore.
- Integrated Elasticsearch for hotel search, filtering and availability workflows.
- Built Kibana monitoring and CI/CD delivery with Docker, Kubernetes and Google Cloud.

PLATFORM WORKFLOW

REQUEST	>	SERVE	>	STORE	>	SEARCH	>	OPERATE
booking		Go APIs		PostgreSQL		Elastic		GCP/K8s

DELIVERY PROFILE

50,000+	REST	CI/CD
events per day	**service contracts**	**cloud delivery**
reliable · testable · traceable	reliable · testable · traceable	reliable · testable · traceable

02 Commerce

StoreFront E Commerce Platform

Technologies: React.js · Node.js · PHP · MySQL · JavaScript · MVC

Impact: Built a complete commerce application spanning user flows, backend logic, database operations and testing.

- Implemented a full-stack commerce application with React, Node.js and PHP backend services.
- Connected product, user and checkout workflows to a MySQL data layer.
- Used MVC structure, prototyping and repeatable testing to keep application logic maintainable.

PLATFORM WORKFLOW

BROWSE	>	PROCESS	>	STORE	>	CHECKOUT	>	VERIFY
React UI		Node/PHP		MySQL		orders		tests

DELIVERY PROFILE

FULL STACK

user to database

reliable · testable · traceable

MVC

maintainable code

reliable · testable · traceable

TEST

repeatable delivery

reliable · testable · traceable

03 **Testing and Quality**

Python Web Applications and Automated Validation

Technologies: Python · Django · Flask · PostgreSQL · Pytest · Docker

Impact: Maintained 5+ applications serving around 2,000 daily users and reduced the QA cycle by 25 percent.

- Built and maintained Python web applications and REST APIs with Django, Flask and PostgreSQL.
- Automated validation with Pytest and reusable backend test components.
- Used Docker and Git-based workflows to support consistent engineering delivery.

PLATFORM WORKFLOW

REQUEST	>	VALIDATE	>	STORE	>	TEST	>	RELEASE
web app		API		PostgreSQL		Pytest		CI/CD

DELIVERY PROFILE

5+	2,000	25%
applications	**daily users**	**faster QA cycle**
reliable · testable · traceable	reliable · testable · traceable	reliable · testable · traceable

RELIABILITY AND SCALE

04 Monitoring

Real-Time Vehicle Fault and Emission Anomaly Detection Platform

Technologies: Python · FastAPI · PostgreSQL · React · WebSocket · Monitoring

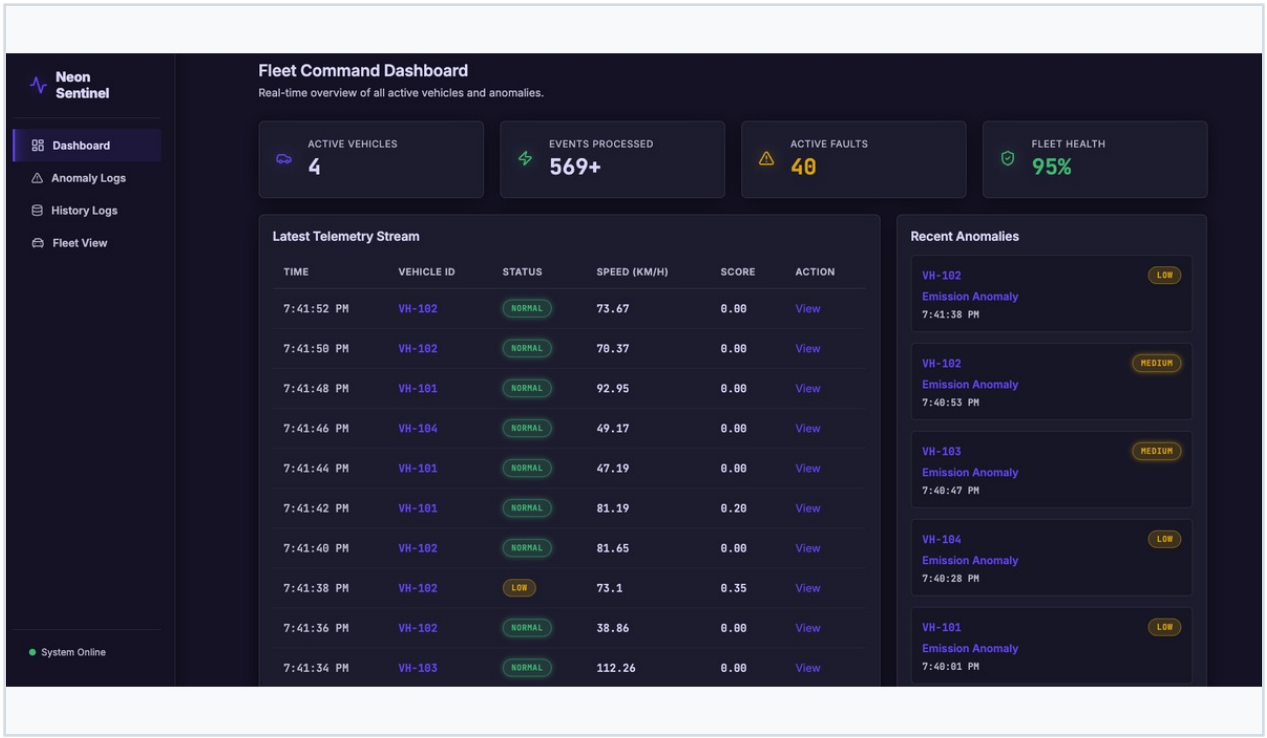
Impact: Built an end-to-end telemetry system with structured alerts, operational KPIs and root-cause visibility.

- Streamed and validated vehicle telemetry through backend services and PostgreSQL.
- Built live dashboards with React and WebSocket updates.
- Designed alerts and investigation views for abnormal system behavior.

PLATFORM WORKFLOW

STREAM telemetry > **INGEST** FastAPI > **STORE** PostgreSQL > **DETECT** anomaly > **DISPLAY** live UI

WORKING PRODUCT



Live fleet monitoring and anomaly investigation dashboard

05 **Large Data**

Euro 7 Automotive Data and ML Pipeline

Technologies: Python · Pandas · NumPy · scikit-learn · Autoencoders · PCA · UMAP

Impact: Processed more than 25 GB and approximately 90 million OBD and CAN time-series records.

- Built modular processing steps for cleaning, transformation, feature preparation and validation.
- Developed anomaly detection and clustering workflows for noisy real-world engineering data.
- Produced traceable technical findings and test specifications for engineering stakeholders.

PLATFORM WORKFLOW

COLLECT OBD/CAN > **CLEAN** validate > **FEATURE** transform > **MODEL** anomaly > **REPORT** findings

DELIVERY PROFILE

25+ GB	90M	ML
source data	**time-series rows**	**anomaly workflow**
reliable · testable · traceable	reliable · testable · traceable	reliable · testable · traceable

06 **API and Data Integrity**

ClaimVision AI Multimodal Insurance Claim Analyzer

Technologies: Python · FastAPI · PostgreSQL · OCR · Structured Extraction · Validation

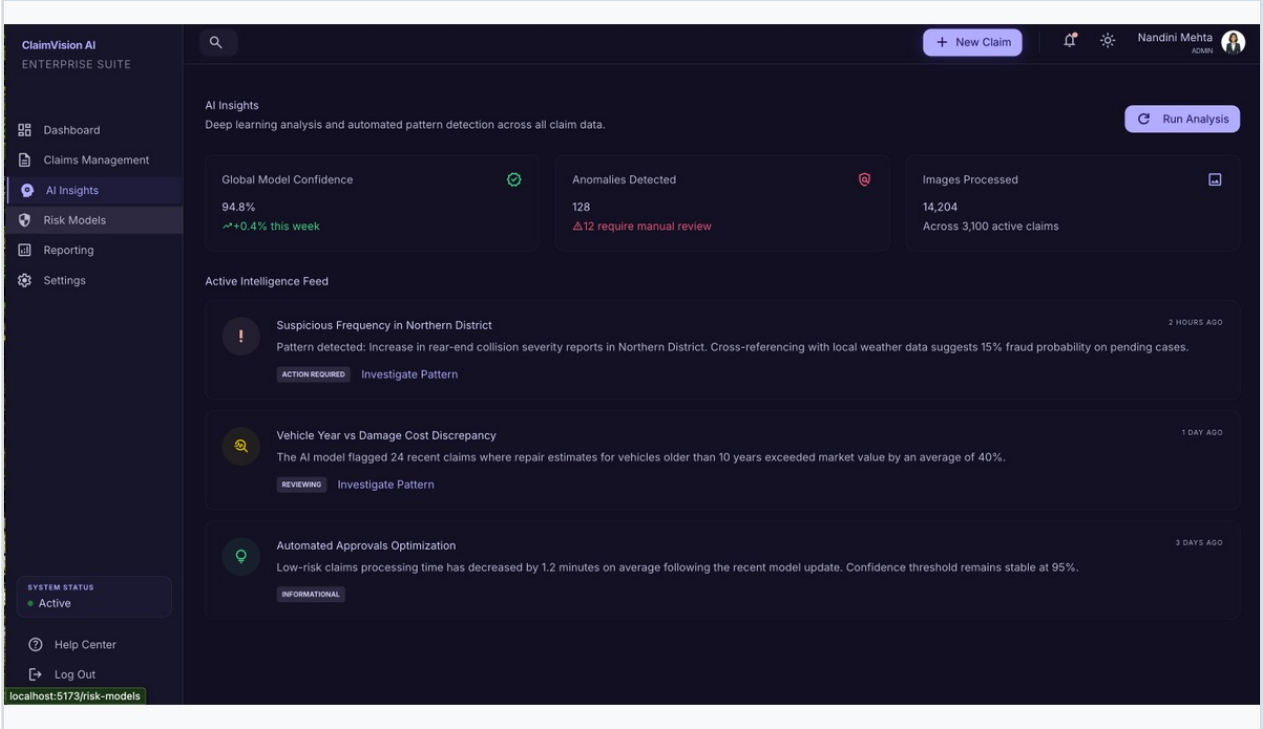
Impact: Designed a modular processing system where multiple input types produce consistent structured outputs.

- Combined claim documents, repair invoices and vehicle damage images in one workflow.
- Added extracted values, consistency checks and risk indicators with validation across inputs.
- Separated backend processing components so they could be tested and evolved independently.

PLATFORM WORKFLOW

UPLOAD	>	EXTRACT	>	NORMALIZE	>	ASSESS	>	DELIVER
docs/images		OCR/vision		schema		risk		report

WORKING PRODUCT



Multimodal claim intelligence and automated pattern detection

07 **Data Processing**

Document Classification and Extraction

Technologies: Python · FastAPI · Tesseract · Hugging Face · NLP · Structured Outputs

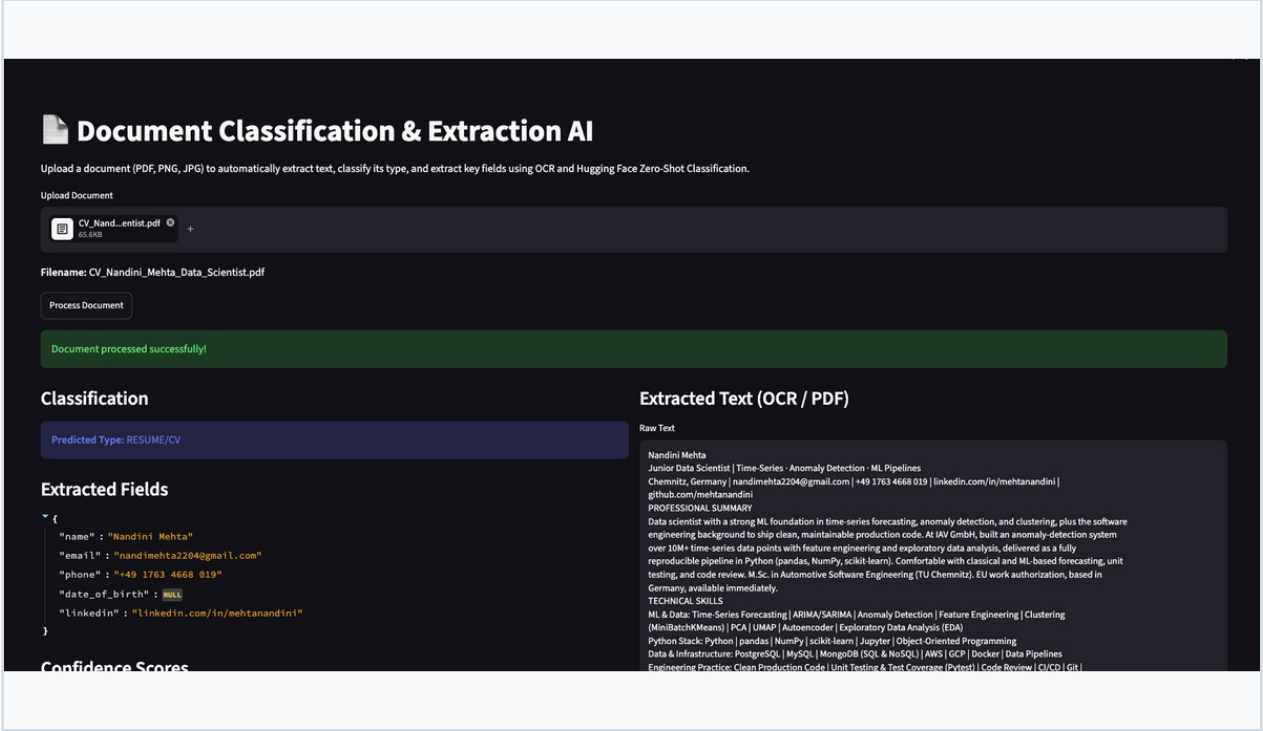
Impact: Built a reusable document ingestion and extraction workflow with validation at the API boundary.

- Processed PDF and image documents through OCR, classification and field extraction steps.
- Added FastAPI interfaces with input validation and structured response formats.
- Used repeatable test inputs to debug extraction quality and data consistency.

PLATFORM WORKFLOW

UPLOAD PDF/image > **READ** OCR > **CLASSIFY** document > **EXTRACT** fields > **RETURN** JSON/API

WORKING PRODUCT



Document classification, OCR text and structured field extraction

08 Document Intelligence and Agent-Ready Workflows

CaseLedger AI

Technologies: Python · FastAPI · SQLAlchemy · PostgreSQL · HTML/CSS/JavaScript · Pytest · Docker · GitHub Actions

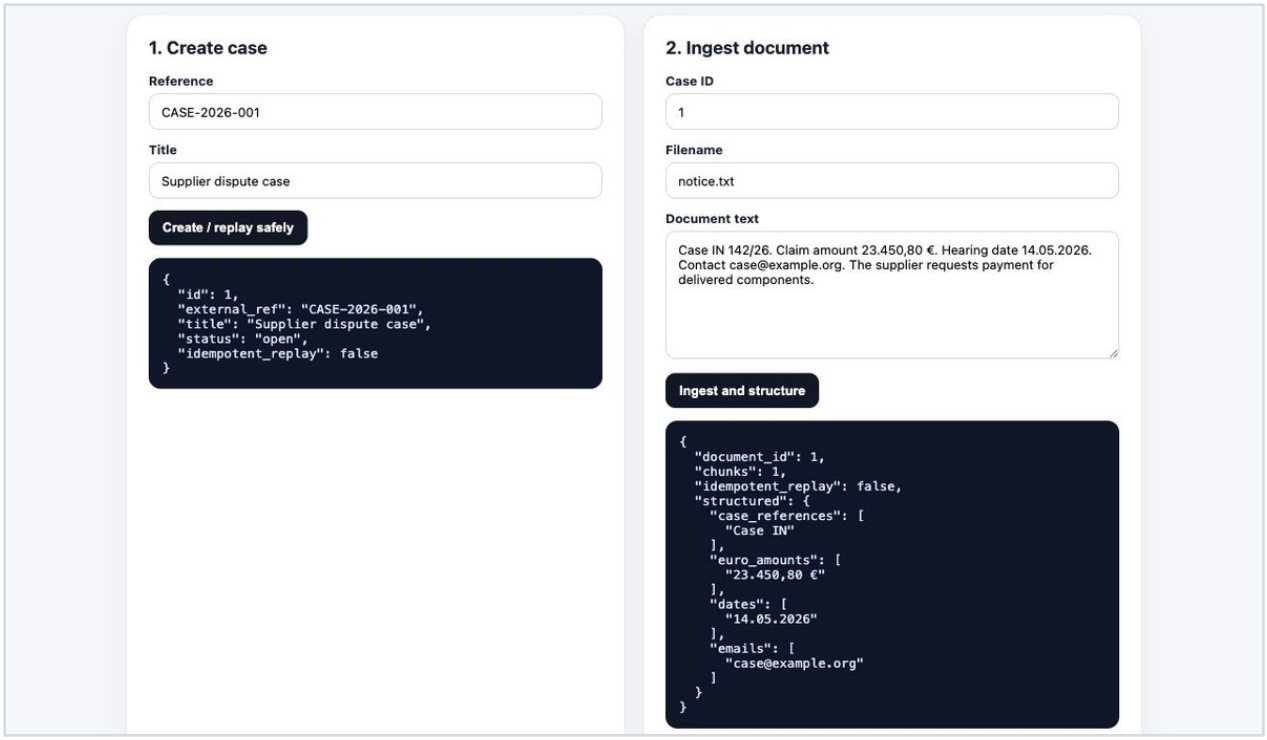
Impact: Built a tenant-isolated document workflow for regulated case files with evidence-backed retrieval, human review and auditability.

- Designed tenant-scoped case and document models with cross-tenant protection and idempotent imports.
- Built structured extraction and grounded question answering with supporting evidence.
- Added audit trails, validation, API tests and deterministic local processing.

PLATFORM WORKFLOW

CREATE case > **INGEST** document > **EXTRACT** structure > **RETRIEVE** evidence > **REVIEW** audit

WORKING PRODUCT



Grounded answer, supporting evidence and auditable case workflow

09 **Enterprise AI Delivery and Governance**

Enterprise AI Delivery Workbench

Technologies: Python · FastAPI · React · TypeScript · ChromaDB · SQLAlchemy · Docker Compose

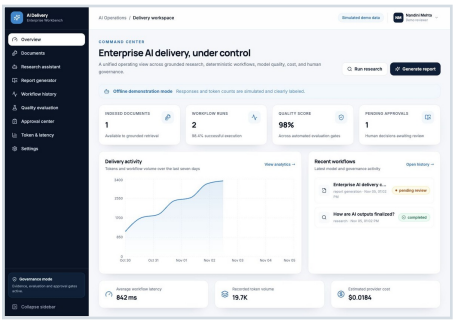
Impact: Built an end-to-end AI delivery platform that keeps enterprise research grounded, measurable and accountable.

- Ingests enterprise documents with isolated retrieval and verified citations.
- Generates schema-validated reports, quality scores and deterministic workflow replays.
- Tracks cost and latency, then records human approval decisions in an audit trail.

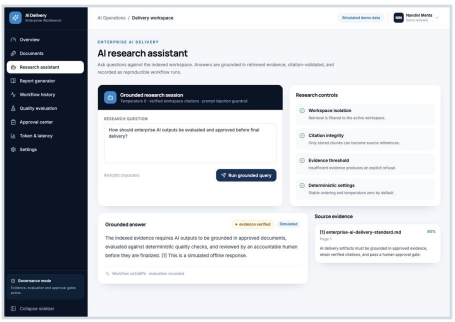
PLATFORM WORKFLOW

INGEST documents > **RETRIEVE** evidence > **GENERATE** reports > **EVALUATE** quality > **APPROVE** audit

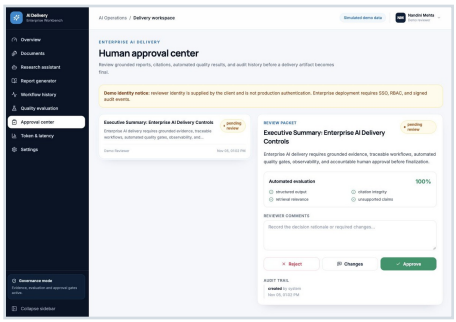
WORKING PRODUCT



Command center



Grounded research



Human approval

Governed delivery from evidence to accountable approval

Grounded RAG · structured reports · automated evaluation · observability · human audit trail

10 Machine Learning and Energy Systems

EnergyML Research Platform

Technologies: Python | FastAPI | scikit-learn | PostgreSQL | SQLAlchemy | Docker Compose

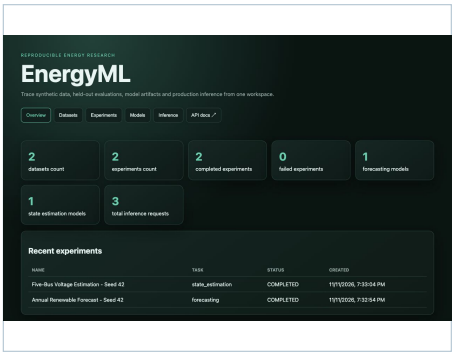
Impact: Built a reproducible research platform from deterministic energy data through held-out model evaluation, artifact verification and validated inference.

- Generated 8,760 hourly renewable observations and 5,000 five-bus network samples.
- Reduced forecasting MAE by 37.6 percent versus a 24-hour persistence baseline.
- Achieved 0.000269 p.u. voltage-estimation MAE with hashed artifacts and API inference.

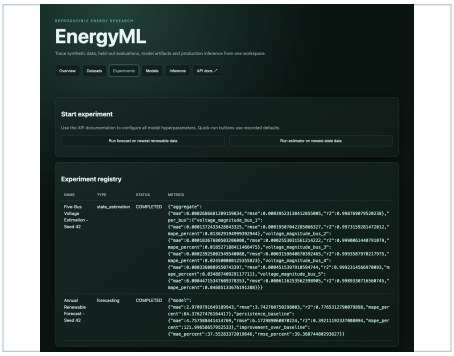
PLATFORM WORKFLOW

GENERATE seeded data > ENGINEER safe features > TRAIN ML pipelines > VERIFY metrics + hashes > SERVE FastAPI

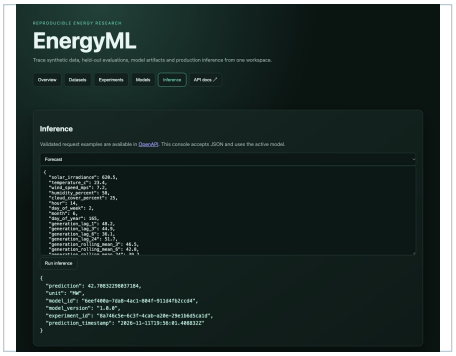
WORKING PRODUCT



Overview and registry counts



Completed experiments



Active-model inference

Reproducible research from seeded data to active-model inference

2 datasets | 2 completed experiments | 2 active models | 19 tests | Docker/PostgreSQL

CORE ENGINEERING CAPABILITIES

Software engineering	Go, Python, Java and PHP experience across APIs, microservices, web backends and full stack applications.
Data and databases	Professional PostgreSQL experience plus MySQL, Oracle, Firebase, Elasticsearch and large scale data processing.
Automated testing	Unit and integration testing plus Pytest automation that reduced a QA cycle by 25 percent.
Scalable systems	Services processing more than 50,000 events per day plus a data pipeline over approximately 90 million records.
Messaging and events	Worked with Google Cloud Pub/Sub, MQTT and event driven application components.
Reliability and observability	Elasticsearch, Kibana, Grafana, monitoring, debugging, alerts and service health investigation.
Cloud delivery	Docker, Kubernetes, Google Cloud, Git based workflows and automated CI/CD delivery.
Product development	Built ecommerce, booking, document intelligence, AI and real time monitoring applications across the stack.
AI and agentic systems	LLM APIs, RAG, LangGraph, Hugging Face, multimodal AI, OCR, NLP, semantic search, vector databases and model evaluation.

ADDITIONAL ENGINEERING PROJECT

Wildlife Protecting System | Bachelor Thesis

Built an embedded railway safety system using Java, Arduino and PIR sensors with MQTT and HTTP based alerting. The project strengthened my foundations in event driven communication, real time monitoring and system reliability.

PROFESSIONAL OUTCOMES

50,000+

events per day supported by backend and cloud services

90 million

automotive records processed in a large data pipeline

2,000

daily users across Python web applications

25 percent

reduction in QA cycle through Pytest automation

ENGINEERING STRENGTHS

- Design APIs and software components around clear data contracts and traceable behaviour.
- Treat automated testing, monitoring and CI/CD as part of feature delivery from the beginning.
- Use databases, validation and observability to protect data consistency and diagnose failures quickly.
- Combine backend, full stack, data, AI and agentic systems experience with event driven cloud development.